

```
[3]: # Train Seat Reservation System using List

# Create 20 seats initialized to 0
# 0 = Available
# 1 = Booked

seats = [0] * 20

while True:
    print("\n--- Train Seat Reservation System ---")
    print("1. Book Seat")
    print("2. Cancel Seat")
    print("3. Show Seats")
    print("4. Booking Percentage")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    # Book a seat
    if choice == 1:
        seat_no = int(input("Enter seat number to book (1-20): "))

        if seat_no < 1 or seat_no > 20:
            print("Invalid seat number!")

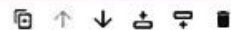
        elif seats[seat_no - 1] == 1:
            print("Seat already booked!")

        else:
            seats[seat_no - 1] = 1
            print("Seat", seat_no, "booked successfully.")

    # Cancel a seat
    elif choice == 2:
        seat_no = int(input("Enter seat number to cancel (1-20): "))

        if seat_no < 1 or seat_no > 20:
            print("Invalid seat number!")

        elif seats[seat_no - 1] == 0:
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        elif seats[seat_no - 1] == 0:
            print("Seat is already available!")

        else:
            seats[seat_no - 1] = 0
            print("Seat", seat_no, "cancelled successfully.")

# Display seats
elif choice == 3:
    print("\nSeat Status:")
    for i in range(20):
        status = "Booked" if seats[i] == 1 else "Available"
        print("Seat", i + 1, ":", status)

# Display booking percentage
elif choice == 4:
    booked = seats.count(1)
    percentage = (booked / 20) * 100

    print("Total Booked Seats:", booked)
    print("Booking Percentage:", percentage, "%")

# Exit program
elif choice == 5:
    print("Thank you for using the system.")
    break

else:
    print("Invalid choice! Please try again.")

```

--- Train Seat Reservation System ---

```

1. Book Seat
2. Cancel Seat
3. Show Seats
4. Booking Percentage
5. Exit
Enter your choice: 3

```

```

Seat Status:
Seat 1 : Available
Seat 2 : Available
Seat 3 : Available

```

```
4. Booking Percentage
5. Exit
Enter your choice: 5
Thank you for using the system.
```

```
[4]: # Python Program to Analyze a YouTube Playlist using Lists

# Store video durations in a list
videos = [12, 4, 8, 15, 3, 20]

# Total watch time
total_time = sum(videos)

# Longest video
longest_video = max(videos)

# Average video duration
average_duration = total_time / len(videos)

# Videos shorter than 5 minutes
short_videos = []

for video in videos:
    if video < 5:
        short_videos.append(video)

# Display results
print("Video Durations:", videos)

print("Total Watch Time:", total_time, "minutes")

print("Longest Video:", longest_video, "minutes")

print("Average Video Duration:", average_duration, "minutes")

print("Videos Shorter Than 5 Minutes:", short_videos)

Video Durations: [12, 4, 8, 15, 3, 20]
Total Watch Time: 62 minutes
Longest Video: 20 minutes
Average Video Duration: 10.333333333333334 minutes
Videos Shorter Than 5 Minutes: [4, 3]

[5]: # Python Program for Social Media Hashtag Tracker
```

```
[5]: # Python Program for Social Media Hashtag Tracker

# Store hashtags in a list
hashtags = ["#fun", "#python", "#fun", "#code", "#python"]

# Find frequency of each hashtag
frequency = {}

for tag in hashtags:
    if tag in frequency:
        frequency[tag] += 1
    else:
        frequency[tag] = 1

# Find most repeated hashtag
most_repeated = max(frequency, key=frequency.get)

# Find unique hashtags
unique_hashtags = list(set(hashtags))

# Remove duplicate hashtags
removed_duplicates = []

for tag in hashtags:
    if tag not in removed_duplicates:
        removed_duplicates.append(tag)

# Display results
print("Hashtags List:", hashtags)

print("\nFrequency of Each Hashtag:")
for tag, count in frequency.items():
    print(tag, ":", count)

print("\nMost Repeated Hashtag:", most_repeated)

print("\nUnique Hashtags:", unique_hashtags)

print("\nList After Removing Duplicates:", removed_duplicates)

Hashtags List: ['#fun', '#python', '#fun', '#code', '#python']
```

Frequency of Each Hashtag:

#fun : 2

#python : 2

#code : 1

Most Repeated Hashtag: #fun

Unique Hashtags: ['#fun', '#code', '#python']

List After Removing Duplicates: ['#fun', '#python', '#code']

[6]: # Python Program for Expense Splitter among Friends

Store expenses in a list

expenses = [1200, 850, 1500, 950]

Calculate total expense

total_expense = sum(expenses)

Calculate average share

average_share = total_expense / len(expenses)

Find highest expense paid

highest_paid = max(expenses)

Find person who paid highest

person_number = expenses.index(highest_paid) + 1

Display results

print("Expenses Paid by Friends:", expenses)

print("\nTotal Expense:", total_expense)

print("Average Share per Friend:", average_share)

print("Highest Amount Paid:", highest_paid)

print("Person", person_number, "paid the highest amount.")

print("\nEqual Contribution Details:")

for i in range(len(expenses)):

 difference = expenses[i] - average_share

```
if difference > 0:
    print("Person", i + 1, "should receive", difference)
elif difference < 0:
    print("Person", i + 1, "should pay", abs(difference))
else:
    print("Person", i + 1, "paid exact share.")
```

Expenses Paid by Friends: [1200, 850, 1500, 950]

Total Expense: 4500

Average Share per Friend: 1125.0

Highest Amount Paid: 1500

Person 3 paid the highest amount.

Equal Contribution Details:

Person 1 should receive 75.0

Person 2 should pay 275.0

Person 3 should receive 375.0

Person 4 should pay 175.0

[7]: # Python Program for Pattern Compression

Input list

```
numbers = [1, 1, 1, 2, 2, 3, 3, 3]
```

Empty list to store compressed result

```
compressed = []
```

Initialize count

```
count = 1
```

Loop through list

```
for i in range(len(numbers) - 1):
```

Check if current element is same as next element

```
if numbers[i] == numbers[i + 1]:
```

```
    count += 1
```

```
else:
```

Store element and count as tuple

```
    compressed.append((numbers[i], count))
```

Reset count

```
    count = 1
```

```
# Reset count
count = 1
```

```
# Add last element
compressed.append((numbers[-1], count))
```

```
# Display output
print("Input List :", numbers)
print("Compressed Output :", compressed)
```

```
Input List : [1, 1, 1, 2, 2, 3, 3, 3]
Compressed Output : [(1, 3), (2, 2), (3, 3)]
```

```
[8]: # Mini Shopping Cart System using Lists
```

```
# Empty lists for items and prices
cart = []
prices = []
```

```
while True:
    print("\n--- Mini Shopping Cart System ---")
    print("1. Add Item")
    print("2. Remove Item")
    print("3. Search Item")
    print("4. Calculate Total Bill")
    print("5. Display Duplicate Items")
    print("6. Show Cart")
    print("7. Exit")

    choice = int(input("Enter your choice: "))

    # Add item
    if choice == 1:
        item = input("Enter item name: ")
        price = float(input("Enter item price: "))

        cart.append(item)
        prices.append(price)

        print(item, "added to cart.")

    # Remove item
    elif choice == 2:
        item = input("Enter item name to remove: ")
```



```
elif choice == 2:
    item = input("Enter item name to remove: ")

    if item in cart:
        index = cart.index(item)

        cart.pop(index)
        prices.pop(index)

        print(item, "removed from cart.")

    else:
        print("Item not found in cart.")

# Search item
elif choice == 3:
    item = input("Enter item name to search: ")

    if item in cart:
        print(item, "is available in cart.")
    else:
        print(item, "not found in cart.")

# Calculate total bill
elif choice == 4:
    total = sum(prices)

    print("Total Bill =", total)

# Display duplicate items
elif choice == 5:
    duplicates = []

    for item in cart:
        if cart.count(item) > 1 and item not in duplicates:
            duplicates.append(item)

    if len(duplicates) > 0:
        print("Duplicate Items:", duplicates)
    else:
        print("No duplicate items found.")

# Show cart
```



```
# Show cart
elif choice == 6:
    print("\nItems in Cart:")
    for i in range(len(cart)):
        print(cart[i], "- Rs.", prices[i])

# Exit
elif choice == 7:
    print("Thank you for shopping!")
    break

else:
    print("Invalid choice! Please try again.")
```

--- Mini Shopping Cart System ---

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit

Enter your choice: 5
No duplicate items found.

--- Mini Shopping Cart System ---

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit

Enter your choice: 6

Items in Cart:

--- Mini Shopping Cart System ---

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit

```

        for i in range(len(cart)):
            print(cart[i], "- Rs.", prices[i])

    # Exit
    elif choice == 7:
        print("Thank you for shopping!")
        break

    else:
        print("Invalid choice! Please try again.")

```

--- Mini Shopping Cart System ---

```

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit
Enter your choice: 5
No duplicate items found.

```

--- Mini Shopping Cart System ---

```

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit
Enter your choice: 6

```

Items in Cart:

--- Mini Shopping Cart System ---

```

1. Add Item
2. Remove Item
3. Search Item
4. Calculate Total Bill
5. Display Duplicate Items
6. Show Cart
7. Exit
Enter your choice: 7
Thank you for shopping!

```